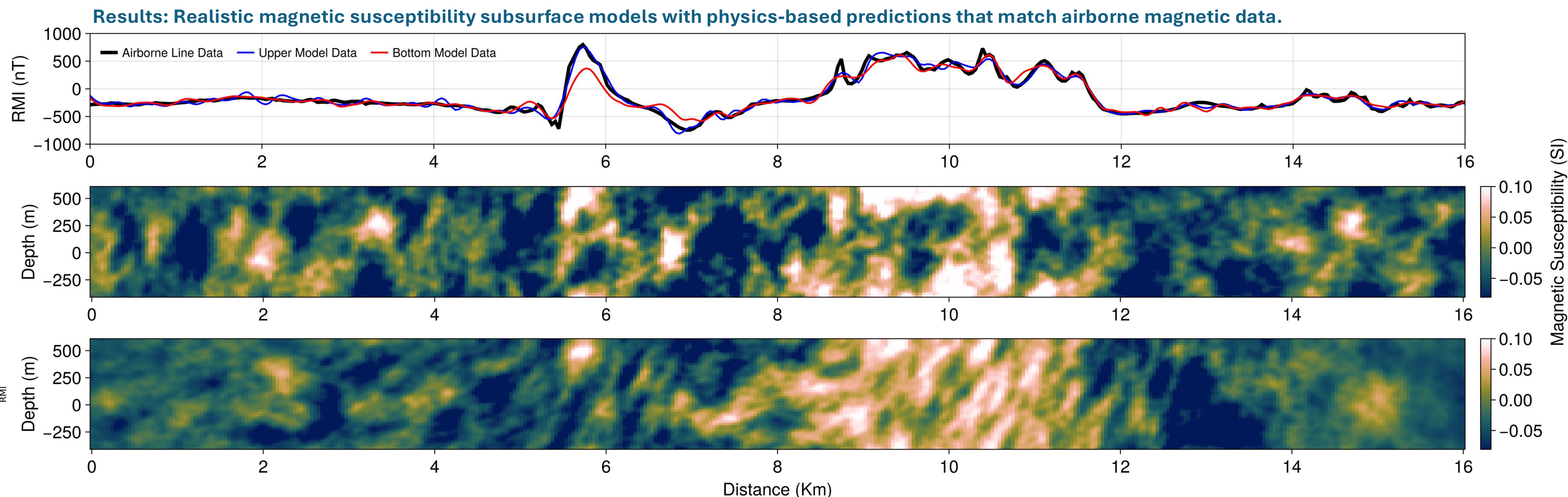
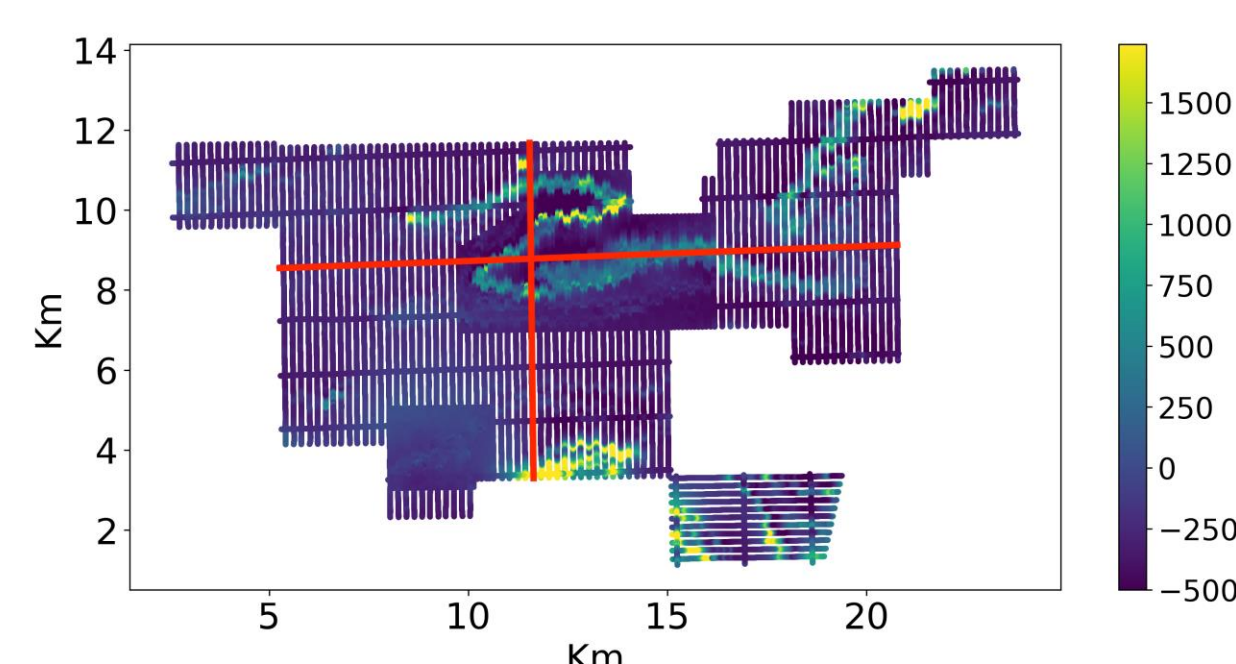


Multi-physics Inversion by Gradual Deformation of Non-Stationary Gaussian Processes

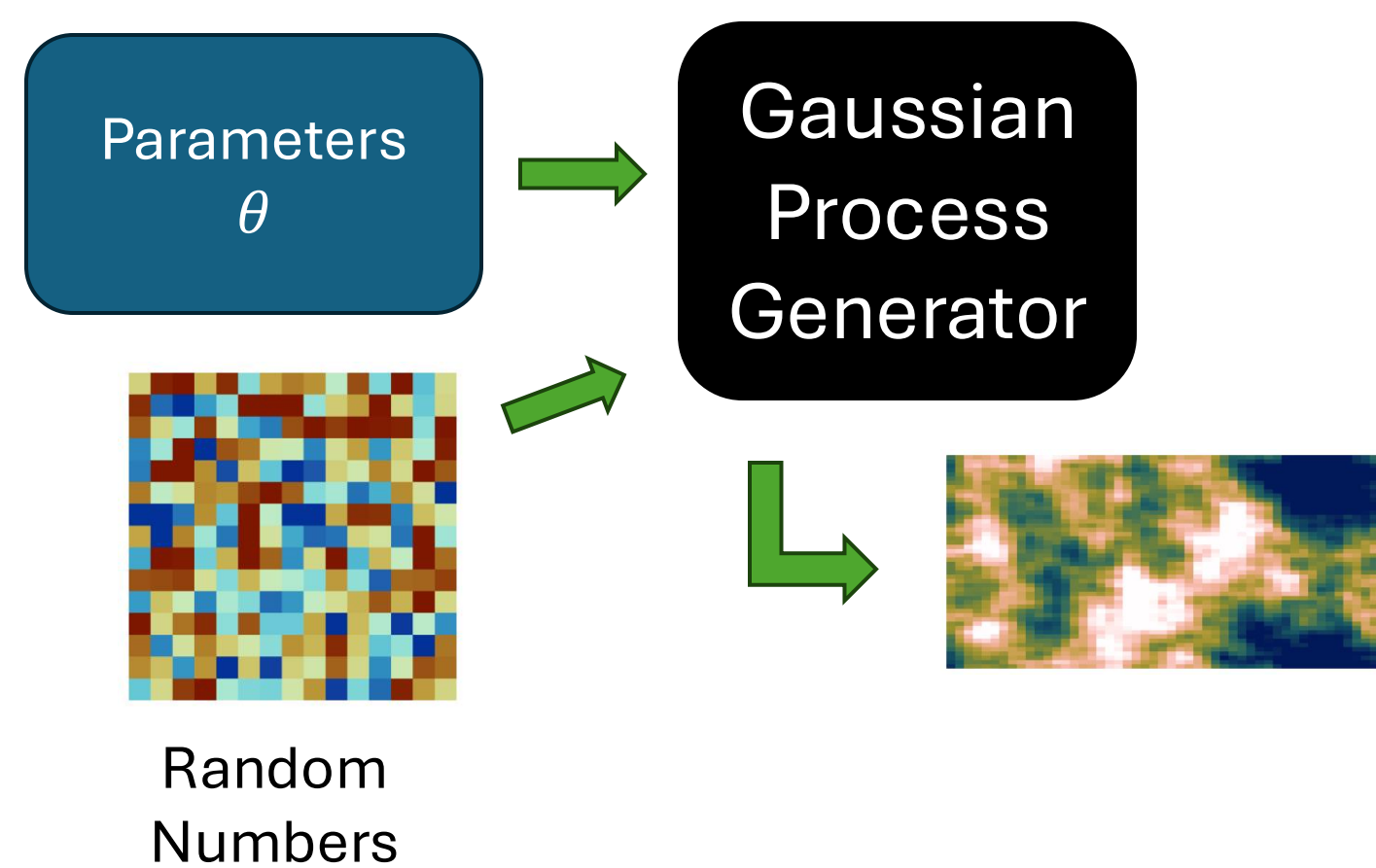
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- We create realistic models, artifact-free, with relevant geological features that match airborne line data over large geographical areas.
- Our models integrate data and physics, and adapt to **multiple** geophysical signals — magnetic data shown here.
- Our approach produces unlimited models to fully quantify uncertainty.
- We quantify uncertainty with numerous realistic models, unlike traditional deterministic models that produce a single scenario.

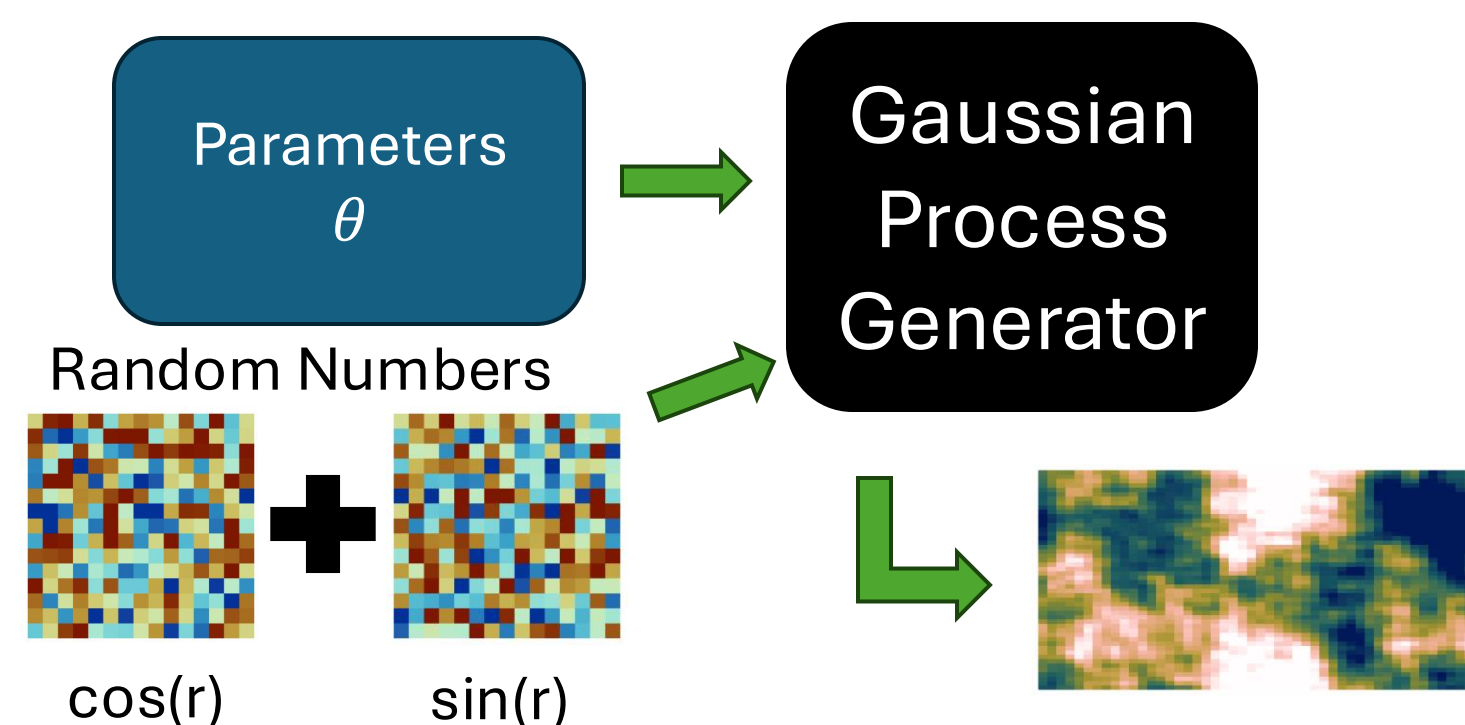
Data from Cape Smith Belt area



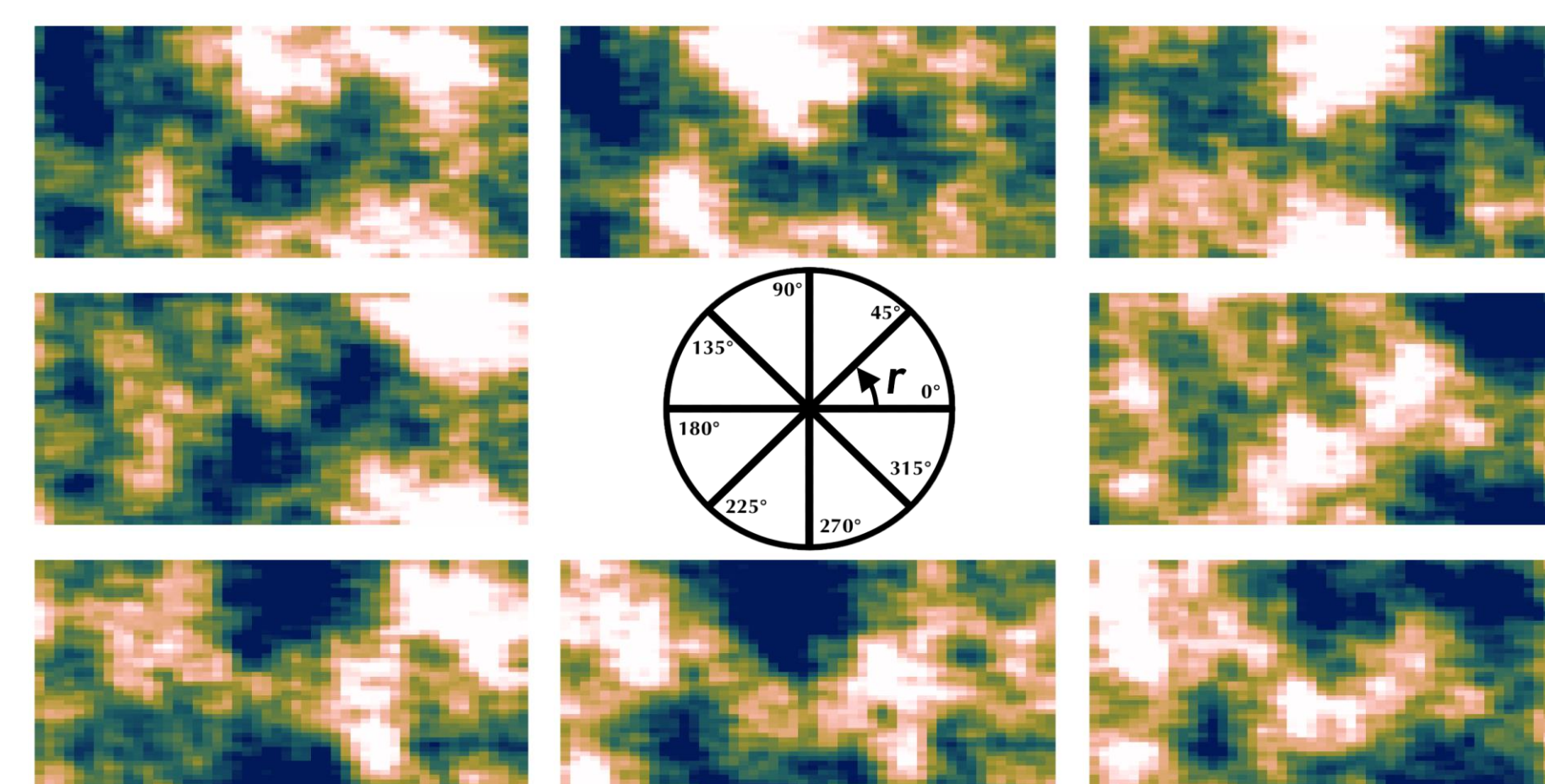
1. Gaussian processes require model parameters and random numbers.



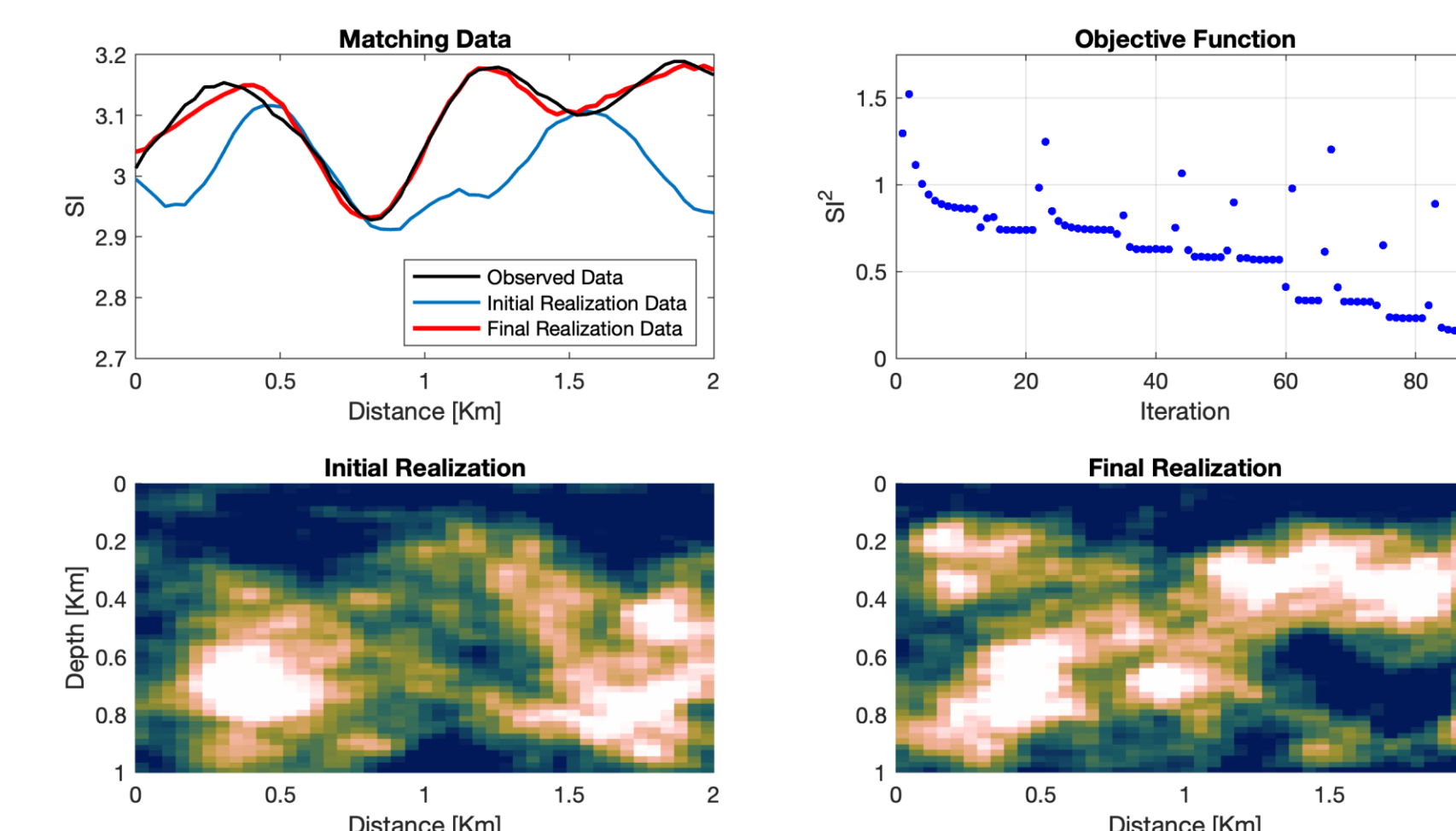
2. We replace the random numbers by a linear combination that preserves variance.



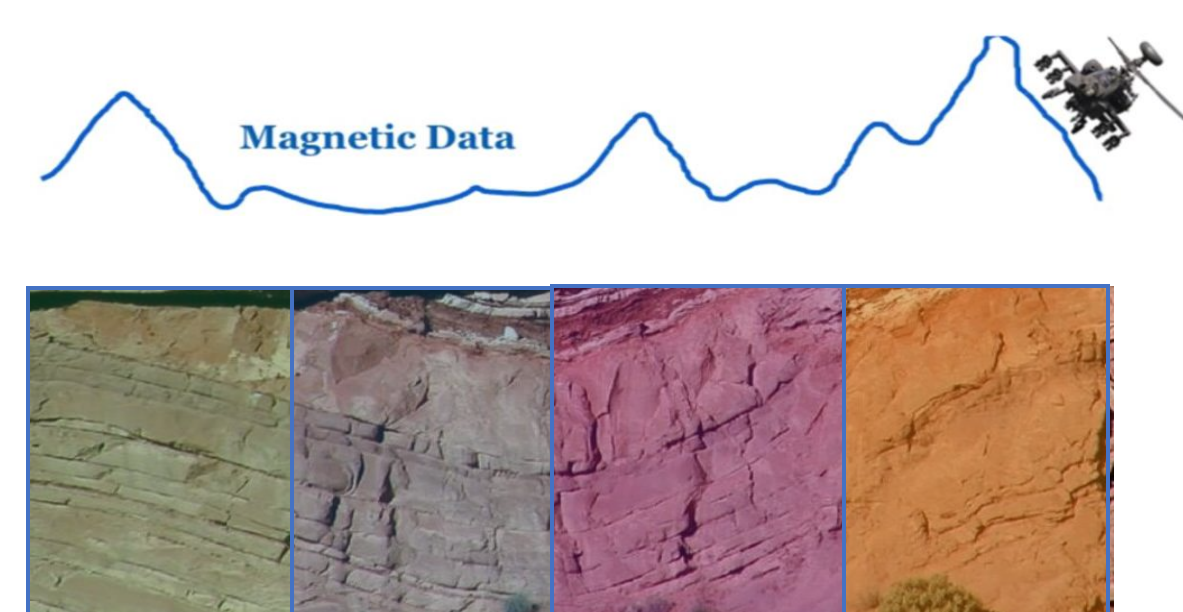
3. We introduce gradual deformation in our Gaussian-based models by varying the parameter 'r'.



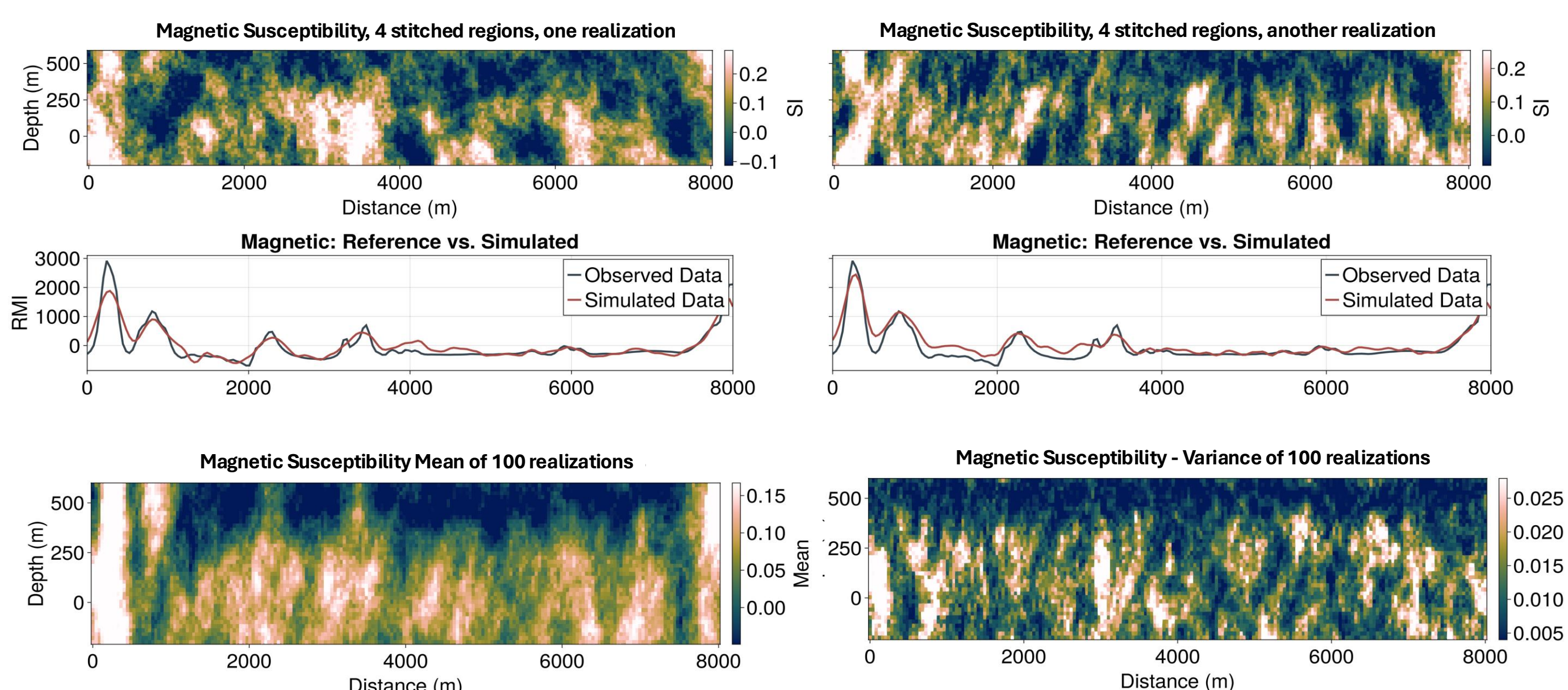
4. We use SimPEG for forward modeling and iteratively optimize 'r' to minimize the error between model and measured data.



5. We capture geological variation by splitting the survey area into regions with different parameters, then stitching them together. Below: two stitched regions; left is constant.



6. We invert one region at a time, stitch the next, invert, and repeat until the full airborne line is inverted. This is repeated hundreds of times to quantify uncertainty.



Theory corner: how parameters for panels are estimated.

Input	Prior $p(\theta_n), p(\varphi_n \theta_n)$
	Geophysical data per panel: $d_{obs,0}, d_{obs,1}, \dots, d_{obs,n}$
Output	Posterior samples $(\varphi_{0,post}, \theta_{0,post}), (\varphi_{1,post}, \theta_{1,post}), \dots, (\varphi_{n,post}, \theta_{n,post})$
Estimate a predictive model for Gaussian process hyperparameters	
Sample: $\theta_j \sim p(\theta_0); \varphi_j \sim p(\varphi_0 \theta_0), j = 1, \dots, N$	
Calculate: $d_j = g(\varphi_j), j = 1, \dots, N$	
Estimate: $\hat{p}(\theta d)$ using samples $(\theta_j, d_j), j = 1, \dots, N$	
Sample $\theta_{0,j}^* \sim p(\theta_0 d_0)$ and $\theta_{1,j}^* \sim p(\theta_{i+1} d_i), j = 1, \dots, N$	
Estimate: $p^*(\theta_1 \theta_0)$ using samples $(\theta_{0,j}^*, \theta_{1,j}^*), j = 1, \dots, N$	
Do Panel 0	
Predict: $\theta_0 \sim \hat{p}(\theta d_{obs,0}); \varphi_0 \sim p(\varphi_0 \theta_0)$	
Filter: $\varphi_{0,post} \sim p(\varphi_0 d_{obs,0}, \theta_0)$ using Gradual Deformation	
For Panel $i = 1, \dots, n$	
Predict $\theta_i^* \sim p^*(\theta_i \theta_{i-1}); \varphi_i \sim p(\varphi_i \theta_{i-1}, \theta_i^*)$ (= Gauss)	
Filter: $\varphi_{i,post} \sim p(\varphi_i d_{obs,i}, \theta_i)$ using Gradual Deformation	
End	Repeat starting with Panel 0